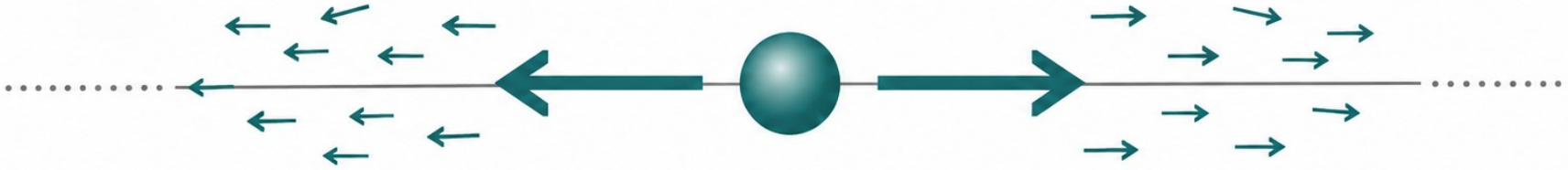
Can Random Motion Produce a Stable Pattern?

If a particle receives random kicks to the left and right with equal likelihood and equal size, will its overall motion show a stable pattern?



1. Equal left/right kicks
2. One path is unpredictable
3. Many paths may have a centre near zero and nonzero spread

Predict:
where will one path finish?
What will many paths show?

Define One Random Trial Before Coding

ONE TRIAL

- One trial = one complete 100-step path

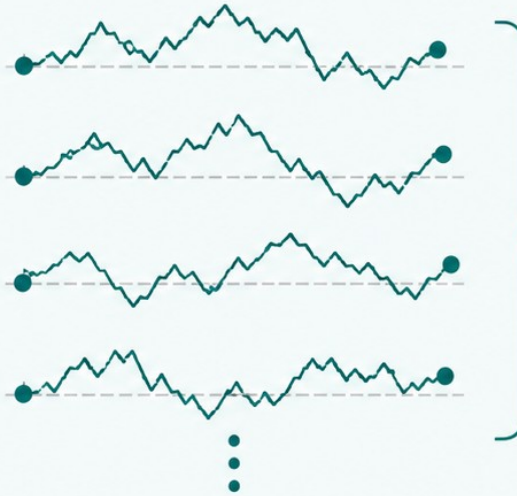


- Start $x = 0$ mm; step length = 1 mm

repeat this complete path

REPEATED TRIALS

- Repeated trials use independent draws



Do not confuse N (steps per path) with M (number of paths).

Read the Random-Step Rule in Physics Language

Each step is +1 mm or -1 mm

+1 mm

−1 mm

equal probability = equal probability

no preferred direction

Equal probability means no preferred direction

N changes the process; M changes the evidence

N

changes the process

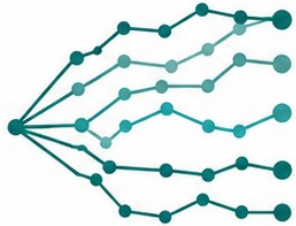
one path of 100 steps



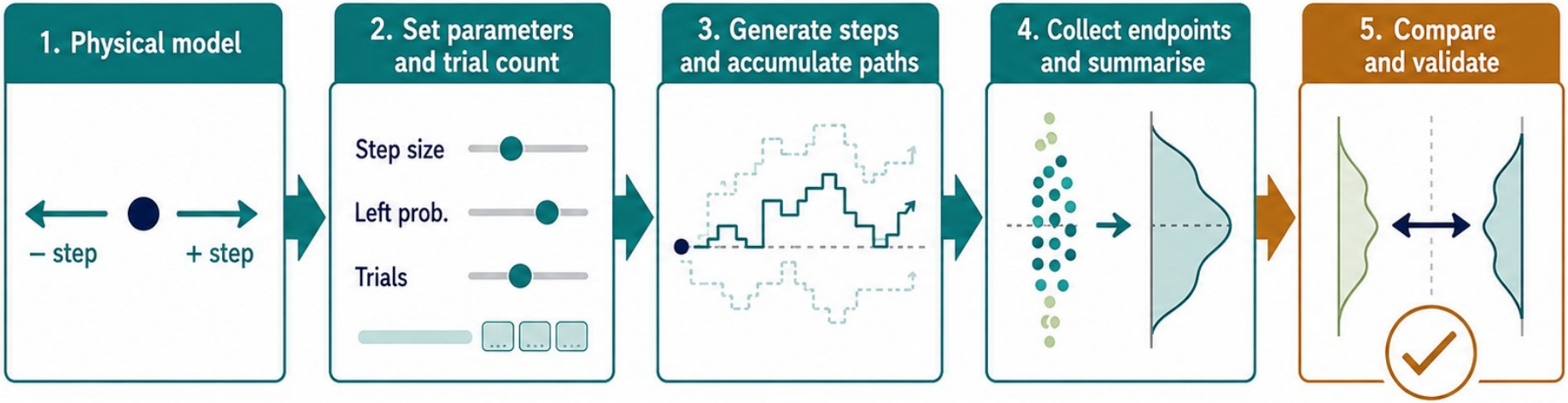
M

changes the evidence

many paths of 100 steps each



Plan the Monte Carlo Experiment



2. Set parameters and trial count
3. Generate steps and accumulate paths
4. Collect endpoints, summarise, compare, validate

Use a Seed for a Reproducible Random Sequence

RUN 1

```
rng(4605,'twister')
```

→



`rng(4605,'twister')` resets a known sequence

=

RUN 2

```
rng(4605,'twister')
```

→

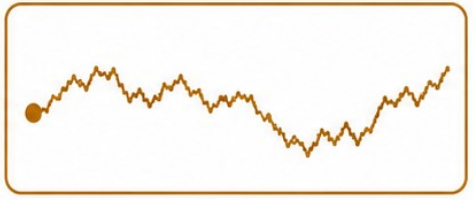


Same seed plus same code repeats results

ANOTHER VALID RUN

```
rng(902,'twister')
```

→



A different seed can still give a valid result

> **!** A seed supports exact reruns — it does not replace validation.

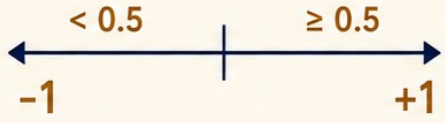
Read the Random-Sampling Scaffold

```
random_draw = rand(n_steps,maximum_trials);
```

→ Uniform random draws

```
step_direction = 2*(random_draw >= 0.5)-1;
```

→ Direction: -1 or +1

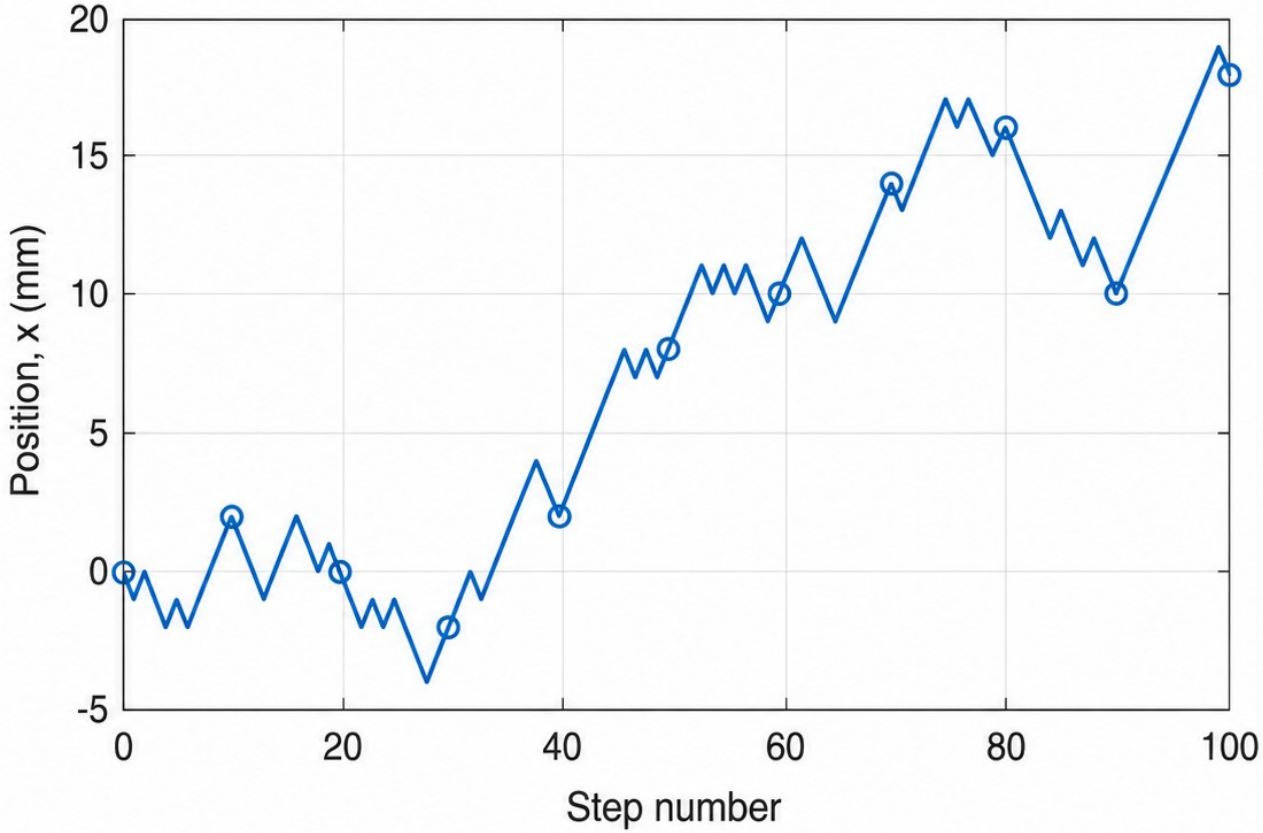



```
step_mm = step_length_mm*step_direction;
```

→ Physical step in mm

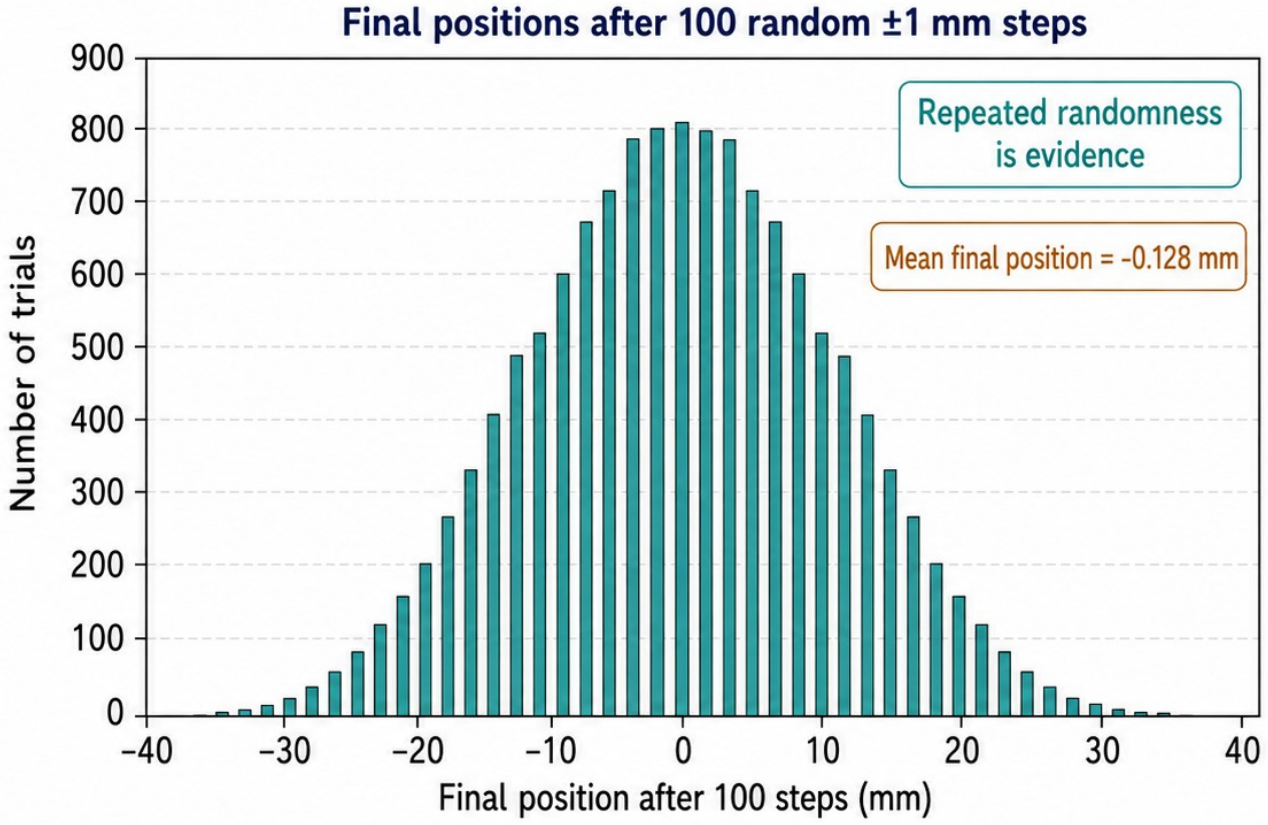
One Trajectory Is One Possible History

- Show one locked 100-step trajectory
- Another seed gives another valid path
- One path cannot determine mean or spread

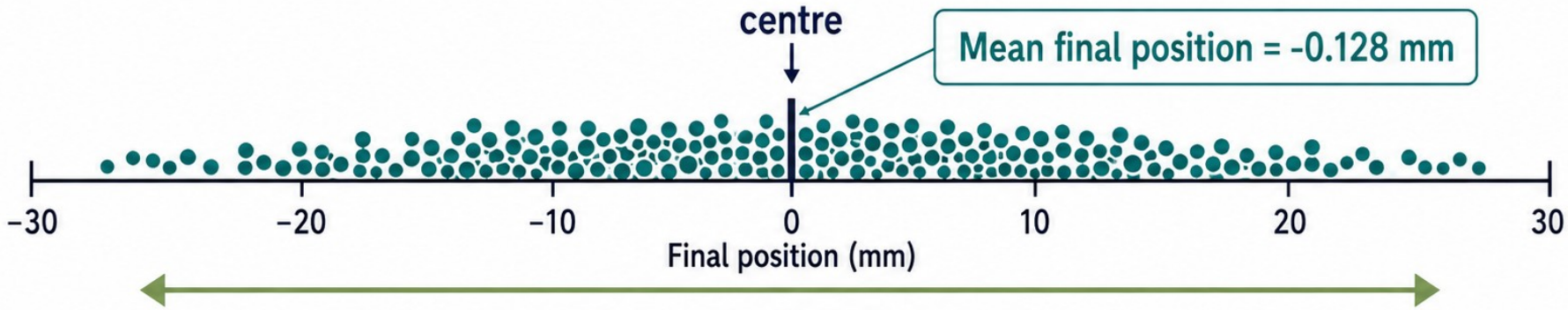


Repeat the Trial to Build a Distribution

1. One trial = one complete 100-step path
2. Repeat 10,000 independent paths
3. The distribution is broad but centred near zero



Mean and Spread Answer Different Questions



Spread = 9.9423 mm

(Trial-to-trial variation in final position)

MEAN

Where is the centre?

SPREAD

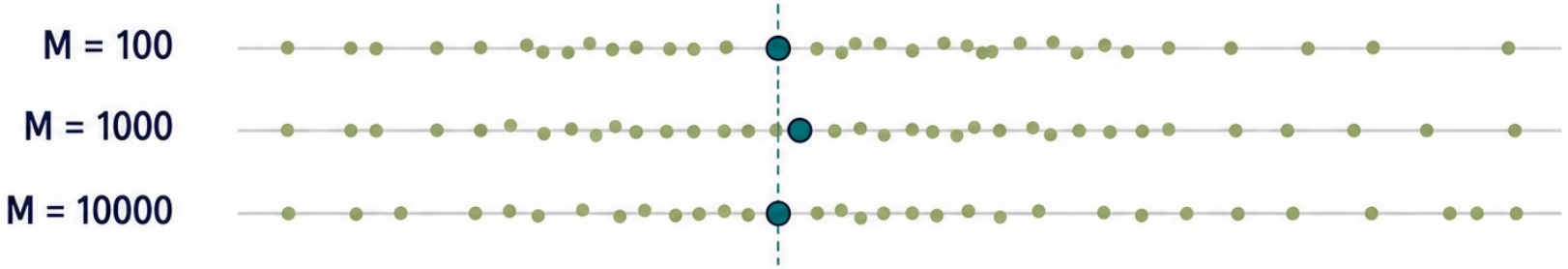
How much do individual outcomes vary?

A mean near zero does not mean every particle ends near zero

More Trials Do Not Remove Physical Variability

M	Mean final position (mm)	Spread (mm)
100	0.040	10.6320
1000	0.066	9.8122
10000	-0.128	9.9423

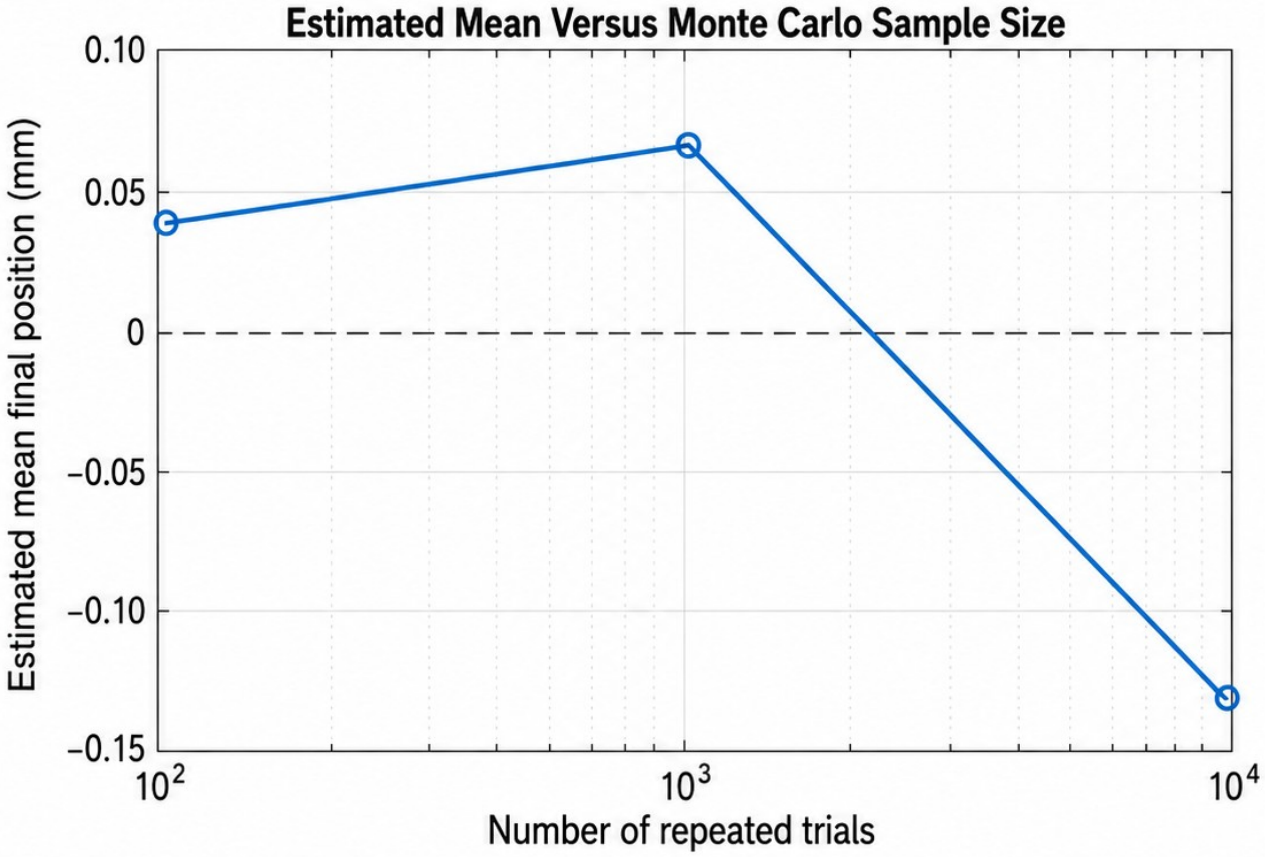
More trials stabilise estimates; they do not remove physical spread.



Monte Carlo Convergence Need Not Be Monotonic

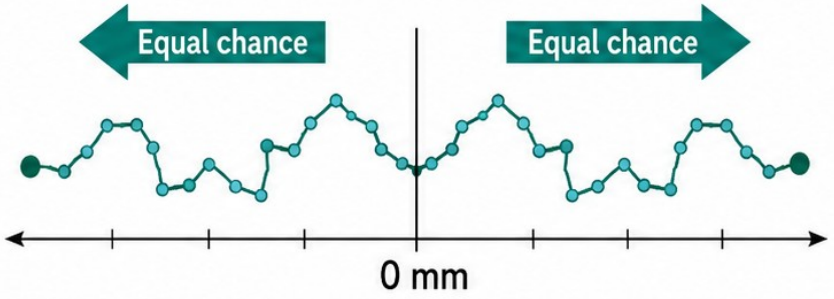
1. Estimated means: 0.040, 0.066, -0.128 mm
2. The 1000-trial result is farther from zero than the 100-trial result
3. More trials generally stabilise; one sequence need not approach monotonically

> ! Non-monotonic behaviour is a normal part of Monte Carlo estimation.



Validate With Symmetry and the Step Rule

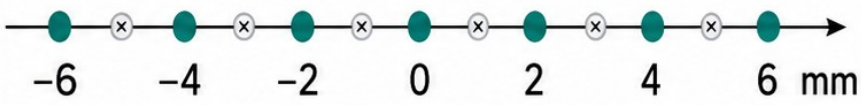
CHECK 1 · SYMMETRY



Symmetry predicts a long-run mean near 0 mm

PASS: mean is near 0 mm

CHECK 2 · STEP RULE



After 100 ±1 mm steps every endpoint is an even integer mm

PASS: endpoints obey the step rule

The locked simulation passes both checks

A plausible histogram or fixed seed is not itself validation.

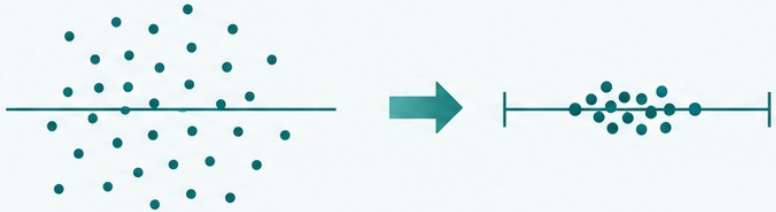
Working Exposure: Precision and Random-Walk Scaling

Working exposure

Precision of the estimated mean

Standard error = spread/sqrt(M)

It falls from 1.0632 to 0.0994 mm



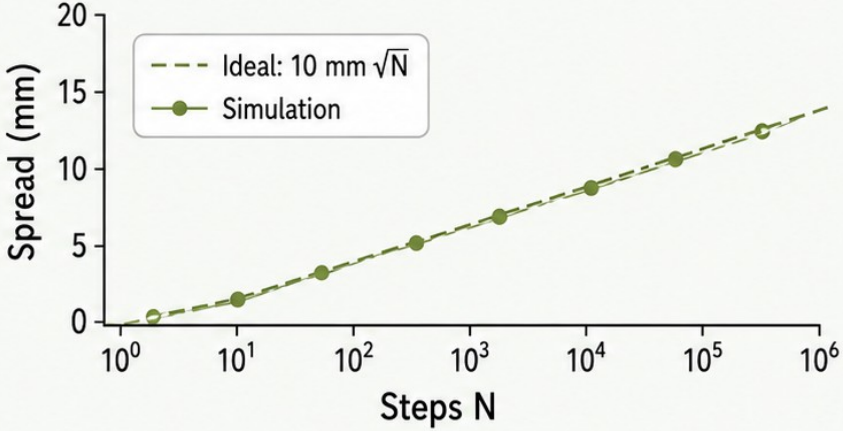
Standard error concerns the estimated mean, not physical spread.

Working exposure

Ideal random-walk scaling

Ideal spread a*sqrt(N) = 10 mm

simulation gives 9.9423 mm



Explain One Trial, One Statistic, and One Check



Define one trial

What exactly happens in one trial?



Explain mean versus spread

What is the mean? What does spread tell you?



Choose and explain one validation check

Which check will you use? What does it verify?



Why can two correct runs differ? When is a fixed seed useful?